Vocard API documentation

Textbook ingestion. Grounded answers. Durable conversations.

Developer reference for the Vocard LangChain backend. Covers the production student and administrator HTTP interfaces, plus the separate local Ollama playground.

Document baseline	Value
Prepared	17 October 2026 (Asia/Seoul)
Project package	vocard-ingestion 0.1.0
Source revision	79d123f; inspected working tree
Primary app	vocard.api:create_app
Local development URL	http://127.0.0.1:8000
Production URL	Deployment-specific; use your HTTPS gateway URL

Reading guide

02 - Endpoint directory and conventions
03 - Authentication and service setup
04 - Student answering
05 - Conversation history
06 - Shared answer contract
07 - Administrator retrieval
08 - Administrator ingestion
09 - Jobs, book history and review
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This reference describes repository behavior, not a verified live deployment. Example identifiers and answer content are illustrative. No external service calls were needed to prepare it.

Endpoint directory

Method and path	Purpose / successful status
GET /healthz	Public process health / 200
POST /ai/answer	Firebase user: ask or continue a lesson chat / 200
GET /ai/conversations	Firebase user: latest chat for exact lesson scope / 200
GET /ai/conversations/{conversation_id}	Firebase user: owned chat / 200
POST /admin/books/{book_id}/ingestions	Administrator: submit PDF and optional extraction / 202
GET /admin/books/{book_id}	Administrator: book and version history / 200
POST /admin/books/{book_id}/search	Administrator: semantic retrieval / 200
POST /admin/books/{book_id}/answer	Administrator: grounded answer / 200
GET /admin/books/{book_id}/lessons	Administrator: exact lesson lookup / 200
GET /admin/jobs/{job_id}	Administrator: job details / 200
POST /admin/jobs/{job_id}/retry	Administrator: retry eligible failed job / 202
GET /admin/jobs/{job_id}/extraction	Administrator: extraction JSON download / 200

Conventions

Paths are relative to the configured base URL. JSON requests use Content-Type: application/json; ingestion uses multipart/form-data. Standard errors use a detail field. Request models for search, answers and local chat reject unknown properties.

Book IDs match [A-Za-z0-9][A-Za-z0-9_-]{0,99}. Student conversation path IDs and request IDs must be UUIDs. PDF page indexes are 1-based and distinct from printed page labels. Database timestamps exposed in book, job and conversation objects are Unix seconds.

Neither app exposes /docs, /redoc or /openapi.json. GET /healthz returns {"status":"ok"}; it is a process response rather than an end-to-end dependency readiness test.

Authentication and startup

Student API: Firebase ID token

Authorization: Bearer <FIREBASE_ID_TOKEN>

The server verifies the Firebase ID token, including revocation, using Firebase Admin and derives the UID from verified claims. Missing, invalid, expired or revoked credentials return 401; anonymous sign-in returns 403. An unconfigured or unavailable verifier returns 503. Client-provided UID and book ID are never accepted by the student question model.

VOCARD_COURSE_BOOKS maps course IDs to authorized book IDs. Every non-anonymous user in the configured Firebase project can access a mapped course. An unmapped course returns 403. There is no per-student enrollment check in this implementation.

Administrator API: separate HS256 JWT

Claim	Requirement
sub	Nonblank administrator identity
iat, exp	Required issued-at and expiry claims
iss	VOCARD_JWT_ISSUER; default vocard-admin
aud	VOCARD_JWT_AUDIENCE; default vocard-ingestion
role	Exactly administrator
book_ids	Optional array of nonblank book IDs; omitted = unrestricted book scope, empty = no books

Use a signing secret of at least 32 bytes. No HTTP token issuance endpoint exists. Missing or invalid admin credentials return 401; wrong role or out-of-scope book returns 403. Admin authentication runs before multipart parsing.

Run the production app and worker separately

```
.venv/bin/uvicorn vocard.api:create_app --factory \
  --host 127.0.0.1 --port 8000
.venv/bin/python -m vocard.worker
```

Export configuration first; the app does not automatically load .env. The API checks private S3 configuration at startup. API and workers share durable SQLite on one host/local volume. Remote access uses an HTTPS reverse proxy. Flutter web requires same-origin routing or gateway CORS configuration.

Student: ask a lesson question

POST /ai/answer - Firebase required

JSON field	Contract
question	Required nonblank string, 1-4,000 characters
conversation_id	Optional UUID or null; omit/null to create a conversation
course_id	Required nonblank string, 1-100 characters
unit_id	Required nonblank string, 1-100 characters
lesson_number	Required strict integer >= 1; use the lesson asset number

```
curl "$BASE_URL/ai/answer" \
  -H "Authorization: Bearer $FIREBASE_ID_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"question":"Explain the rule for this lesson.",
       "conversation_id":null,"course_id":"devils_100",
       "unit_id":"devils-grammar-unit-01","lesson_number":1}'
```

Success: 200 with the shared answer object (page 06), plus a server-created conversation_id string. Pass that UUID with the same course, unit and lesson to continue. Student answers use Korean explanations by default and do not enable generated practice examples. Clients cannot override language, history, book, top_k or fallback settings.

Server-controlled conversation and retrieval

The server maps the course to a book and checks lesson availability before admitting a question. A missing active index returns 404. An existing index with no matching lesson evidence returns insufficient_evidence without calling the generation model. Each follow-up retrieves from the current active index; the last two stored user questions expand short follow-up queries.

Default quota: 50 admitted questions per UID per UTC day, across conversations and restarts. Generation failures consume admitted quota. A conversation has a 15-minute fenced request lease; concurrent questions or a changed lesson scope return 409. Default retrieval may include bounded fallback from other lessons within the fixed course/unit scope.

Client behavior

Store the returned conversation_id. Render answer, status, citations and source examples together. If a request times out, reload history before resubmitting: server processing may finish after disconnect. There is no student request idempotency key or streaming response.

Student: conversation history

GET /ai/conversations - latest chat

Required query parameters: course_id (1-100 characters), unit_id (1-100 characters), lesson_number (integer >= 1). Requires Firebase authentication and a mapped course.

```
curl --get "$BASE_URL/ai/conversations" \
  -H "Authorization: Bearer $FIREBASE_ID_TOKEN" \
  --data-urlencode "course_id=devils_100" \
  --data-urlencode "unit_id=devils-grammar-unit-01" \
  --data-urlencode "lesson_number=1"
```

Returns {"conversation": object} for the most recently updated owned chat in that exact book/course/unit/lesson scope, or {"conversation": null}. This endpoint is not a paginated list of all conversations.

GET /ai/conversations/{conversation_id} - load one chat

Returns the owned conversation object directly, without a conversation wrapper. A nonexistent or another user's UUID returns 404. If the conversation course no longer maps to its original book, the endpoint returns 403.

```
{
  "conversation_id": "<UUID>",
  "book_id": "devils-100",
  "course_id": "devils_100",
  "unit_id": "devils-grammar-unit-01",
  "lesson_number": 1,
  "updated_at": 1792200000.0,
  "turns": [
    {"question": "Explain this rule.", "response": {
      "answer": "...", "status": "answered",
      "citations": [], "source_examples": [],
      "practice_examples": []
    }}
  ]
}
```

The abbreviated stored response above shows field shape only; a grounded answered response normally has validated citations. Stored turn responses do not contain conversation_id. UID and request lease details are excluded from the public history object.

Defaults retain 10 complete exchanges (20 messages) and at most 40,000 serialized characters including metadata. Oldest exchanges are dropped; one oversized turn may not be retained. Only question and bounded answer prose are replayed to the model, not old citations as new evidence. Failed generation does not append a turn.

Shared answer response contract

Field	Meaning
answer: string	Explanation, abstention text, or one focused clarification question
status: string	answered | insufficient_evidence | clarification_needed
citations: array	Verbatim quotes plus server-resolved source locations
source_examples: array	Exact source text and extraction-provided metadata
practice_examples: array	Opt-in generated examples; admin answer only
conversation_id: string	Additional field on POST /ai/answer only

Citation object

```
{
  "chunk_id": "<retrieved-chunk-id>",
  "quote": "<verbatim supporting text>",
  "book_id": "devils-100", "version": 1,
  "index_version": "<immutable-index-id>",
  "source_pages": [15, 16],
  "source_page_labels": [],
  "course_id": "devils_100",
  "unit_id": "devils-grammar-unit-01",
  "lesson_number": 1
}
```

course_id, unit_id and lesson_number may be null. source_page_labels is an array of metadata objects; display printed labels separately from source_pages. The server supplies book/version/page metadata. Quotes and IDs are checked against the exact chunks included in generation context.

Example objects

source_examples items contain chunk_id, text, is_correct (boolean or null), and translations (array copied from extraction). Preserve incorrect examples, original punctuation and correctness markers; do not silently repair textbook content.

practice_examples items contain english, translation and the exact server label below. Display the label alongside each generated example.

새로 만든 연습 예문 (교재 원문 아님)

Status handling

insufficient_evidence is a successful HTTP 200 outcome and has empty evidence/example arrays. A generated answer with no citations also abstains. clarification_needed asks one focused question. Forged citations, altered quotations or invalid source examples produce HTTP 502; provider failures produce 503. An active index change during retrieval/generation produces 409.

Administrator: retrieval and answers

POST /admin/books/{book_id}/search

JSON field	Constraint / default
query	Required string, 1-10,000 characters
top_k (alias k)	Optional strict integer 1-100; configured default 4
current_lesson	Optional strict integer >= 1
current_lesson_title	Optional nonblank string
course_id, unit_id	Optional nonblank strings; exact scope filters
allow_book_fallback	Boolean; default true
max_fallback	Optional strict integer 0-100; configured default 2
min_score	Optional finite number -1 to 1; configured default 0

Returns {"documents": [...]} using LangChain Document model serialization: page_content, metadata, id and type. Metadata carries source identity, lesson information and retrieval_scope. Scores are attached by retrieval; metadata is richer than a fixed minimal DTO.

```
curl "$BASE_URL/admin/books/grammar-book/search" \
  -H "Authorization: Bearer $ADMIN_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"query":"gerund as an object","current_lesson":7,
      "top_k":4,"allow_book_fallback":false}'
```

POST /admin/books/{book_id}/answer

Accepts every search field plus language (ko or en; default ko), include_practice_examples (boolean; default false), and history (default []). History allows at most 20 messages and 40,000 total content characters. Each message has role user or assistant and nonblank content of 1-10,000 characters. Returns the shared AnswerResponse without conversation_id. Admin history is supplied per request and is not persisted as a student chat.

GET /admin/books/{book_id}/lessons

Optional query filters: lesson_number >= 1, nonblank title, course_id, unit_id. limit defaults to 50 (range 1-200); offset defaults to 0 (>= 0). Exact metadata lookup in source order, without query embedding or fuzzy matching.

```
{"documents": [...], "total_chunks": 73, "next_offset": 50}
```

next_offset is null when there is no further page. No filters returns scoped chunks. Both search and answering return 404 when the book has no active index; authorization failures return 403. Fallback is lesson-first and bounded, while book/course/unit authorization remains enforced.

Administrator: PDF ingestion

POST /admin/books/{book_id}/ingestions - 202 Accepted

Multipart field	Contract
file	Required PDF upload; bytes must start with %PDF-
extraction	Optional extraction JSON file
reviewed_extraction	Boolean; default false; true requires extraction file
allow_uncertain	Boolean; default false; deprecated, cannot bypass review
course_id, unit_id	Optional nonblank strings, at most 100 characters
chunk_size	Optional integer > configured overlap and <= 50,000

```
curl "$BASE_URL/admin/books/devils-100/ingestions" \
  -H "Authorization: Bearer $ADMIN_TOKEN" \
  -F "file=@/path/book.pdf;type=application/pdf" \
  -F "extraction=@/path/reviewed.json;type=application/json" \
  -F "reviewed_extraction=true" \
  -F "course_id=devils_100" \
  -F "unit_id=devils-grammar-unit-01"
```

The response is the persisted job object plus created (boolean). Important fields: job_id, book_id, sha256, book_version, status, stage, extraction_status, attempts and max_attempts. lease_token is excluded. Acceptance means durable upload and enqueue, not that the book is searchable. Poll the job endpoint until completion and confirm publication.

Size and validation

Default per-file limit is 104,857,600 bytes (100 MiB) for both PDF and extraction. Empty uploads are rejected. Admin middleware caps the complete request at 2 x max_upload_bytes + 1 MiB (201 MiB by default), including streamed bodies. Body limit failures return 413; individual file/input rejection returns 422. PDF processing limits default to 2,000 pages and 30,000 chunks and are enforced during pipeline processing.

Deduplication and publication

Source versions deduplicate by book and PDF SHA-256. Jobs deduplicate by source version and pipeline fingerprint, including extraction hash, review attestation and settings. Resubmitting identical failed input returns the existing failed job. No Idempotency-Key header is implemented.

A separate worker extracts, chunks, embeds and validates an immutable private index. Only successful fenced publication switches the active pointer. Previous active content remains available during processing or failure. A late older job can be superseded rather than roll back a newer publication.

Administrator: jobs and review

GET /admin/jobs/{job_id}

Returns the job record joined with book identity. All admin job responses omit lease_token. Records are currently returned as dictionaries rather than a declared response model; consumers should tolerate additive fields.

Field group	Included data
Identity	job_id, book_version_id, book_id, book_version, sha256, source_key
Lifecycle	status, stage, attempts, max_attempts, available_at, created_at, updated_at, lease_until, last_error
Pipeline / audit	sequence, pipeline_fingerprint, pipeline_json, requested_by, last_retried_by, publication_result
Extraction / review	extraction_key, extraction_status, book_extraction_status, extracted_key, extracted_sha256, review_status, review_report_json, review_report

status is queued, running, succeeded or failed. stage explains progress and may be review_required. Timestamps are Unix seconds. pipeline_json and review_report_json are serialized JSON strings; review_report is parsed JSON or null. Inspect publication_result to distinguish activation from supersession.

POST /admin/jobs/{job_id}/retry - 202

No request body is required. Only failed jobs without review-required status can retry. An eligible retry requeues the job, records the administrator and grants three further attempts. Non-failed jobs and jobs requiring reviewed JSON return 409. Unknown job IDs return 404; job book scope is always checked.

GET /admin/jobs/{job_id}/extraction

Returns saved extraction bytes as application/json with Content-Disposition: attachment; filename="extraction.json". If the job or extraction is unavailable, returns 404. Export this draft, compare every page to the source PDF, correct it, then resubmit the same PDF with extraction and reviewed_extraction=true.

A reviewed import must match the exact PDF hash and cover every PDF page exactly once in order. Empty reviewed pages need review_disposition="blank"; a wholly empty book cannot publish. Embedded approval claims do not authorize review. Review-required jobs do not automatically retry or spend embedding calls.

GET /admin/books/{book_id}

Returns book_id, active_index_version (nullable), created_at, versions and indexes. versions lists source versions in book_version order; indexes lists immutable index history in sequence order. Version records include source hashes, extraction state and review audit data. Index records include object key/checksum, validation_json, job ID and previous_index_version. Unknown books return 404.

Errors and integration behavior

HTTP status	Meaning and client action
400	Malformed Content-Length on admin request
401	Missing/invalid credentials; sign in or refresh the relevant token
403	Wrong role, out-of-scope book, anonymous user or unavailable course
404	Unknown book/job/chat, missing active index or unavailable extraction
409	Index changed, chat busy/expired/wrong lesson, or retry ineligible; inspect detail before retrying
413	Admin request body exceeds the configured cap
422	Invalid JSON/query/form fields, invalid PDF/input, or rejected extra fields
429	Student daily quota reached; Retry-After: 3600 is supplied
502	Generated answer failed validation; local playground also uses this for local model failures
503	Generation/auth unavailable; local playground also uses this for missing Ollama/content
504	Local playground model timeout

Error payloads

{"detail":"Conversation belongs to another lesson"}

{"detail":[{"loc":["body","question"],
"msg":"Field required","type":"missing"}]}

HTTP/application errors normally use a string detail; FastAPI validation errors use an array. Validation entries can contain additional context. Unhandled infrastructure exceptions may produce a generic server error rather than the application detail contract.

Operational guidance

Treat 200 insufficient_evidence as a valid answer state. After a 409 index change, a fresh request can retrieve current evidence. For busy chats, wait and reload history; for a scope conflict, start or select the correct lesson chat. Do not blindly retry every 409.

Quota resets by UTC calendar day, not device timezone. The 429 Retry-After value is a fixed hint, not the exact reset countdown. A transient generation failure consumes an admitted student request. Retried student requests can generate additional turns; verify history after uncertain connection failures.

Responses are regular JSON, not server-sent events. There are no HTTP APIs for student chat deletion, token minting, job cancellation, book deletion or manual index rollback in this implementation.

Separate local Ollama playground

App factory: vocard.chat:create_chat_app. Default URL: http://127.0.0.1:8001. This is an unauthenticated development app with a host allowlist for localhost, 127.0.0.1 and testserver. Keep it bound to loopback. It does not use the production authenticated vector index.

```
.venv/bin/uvicorn vocard.chat:create_chat_app --factory \
  --host 127.0.0.1 --port 8001
```

Route	Result
GET /	HTML playground UI
GET /api/parts	{"parts":[{"id":str,"title":str,"page_count":int}]}
GET /api/parts/{part_id}	{"id":str,"title":str,"pages":[{"number":int,"text":str}]}
POST /api/chat	{"answer":str}

Chat request

```
curl http://127.0.0.1:8001/api/chat \
  -H "Content-Type: application/json" \
  -d '{"part_id":null,"messages":[
      {"role":"user","content":"Explain gerunds."}]}'
```

part_id is optional/null or a string of at most 100 characters. messages requires 1-21 items; each message has role user or assistant and nonblank content of 1-10,000 characters. Total content is at most 40,000 characters. Messages must alternate starting with user and end with user. Unknown request/message fields are rejected.

When part_id is present, reference excerpts are selected from that part within a 14,000-character budget. The source is an unreviewed OCR draft. Part IDs come from /api/parts (for example intro, part-1 or textbook depending on the source). Unknown IDs return 404; unavailable grammar content returns 503.

Ollama is called at http://127.0.0.1:11434/api/chat with stream=false. VOCARD_CHAT_MODEL defaults to qwen2.5:3b; timeout is 180 seconds. This service returns free-form text with no production citation validation contract. Connection errors return 503, timeouts 504, and other local AI failures 502.

The local chat endpoint receives all conversation messages from its caller and does not persist server-side conversations. Browser session state is separate from the student API conversation database.

Configuration and source map

Environment variable	Default / use
VOCARD_S3_BUCKET	Required private bucket
VOCARD_JWT_SECRET	Required; at least 32 bytes
VOCARD_DB_PATH	state/ingestion.sqlite3
VOCARD_FIREBASE_PROJECT_ID	Unset; required to verify student tokens
VOCARD_COURSE_BOOKS	{}; JSON course-to-book map
VOCARD_AI_DAILY_QUOTA	50 per UID per UTC day
VOCARD_AI_HISTORY_MESSAGES	20; bounded 2-20
VOCARD_AI_HISTORY_CHARS	40000; bounded 1000-40000
VOCARD_TOP_K / MAX_FALLBACK	4 / 2 (both use VOCARD_ prefix)
VOCARD_MIN_SCORE	0
VOCARD_ANSWER_MODEL	gpt-4.1-mini
VOCARD_ANSWER_TIMEOUT	60 seconds per attempt; one provider retry
VOCARD_ANSWER_MAX_TOKENS	3000
VOCARD_ANSWER_CONTEXT_CHARS	60000 serialized context characters
VOCARD_CHUNK_SIZE / CHUNK_OVERLAP	1000 / 150 (both use VOCARD_ prefix)
VOCARD_CHAT_MODEL	qwen2.5:3b; local app only
VOCARD_GRAMMAR_PATH	output/pdf/textbook-extraction/textbook.json

OPENAI_API_KEY enables embeddings and grounded answer generation. AWS credentials use the standard boto3 chain; Firebase Admin uses server Application Default Credentials. Use configured index metadata for course/unit/lesson mapping. Uploading a source alone does not grant course access.

Repository references

Production routes: vocard/api.py and vocard/student_api.py. Authentication: vocard/auth.py and vocard/student_auth.py. Answer models and validation: vocard/answering.py and schema/answer-response.schema.json. Conversation ownership, quota and leases: vocard/conversations.py. Upload, job and publication records: vocard/ingestion.py and vocard/db.py. Retrieval: vocard/retrieval.py and vocard/pipeline.py. Settings: vocard/config.py. Local app: vocard/chat.py and vocard/grammar_browser.py. Operational guide: README.md. Extraction contract: schema/grammar-extraction.schema.json.

Verification basis: routes, Pydantic models, persistence code and repository guidance inspected on 17 October 2026. Default settings are repository defaults and can differ in deployment. Examples use placeholders; replace tokens, IDs, paths and the base URL with your own configured values.